

ANH KHOA NGUYEN

Security Engineer

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SUMMARY

I have expertise in Computer Systems, especially at the intersection of computing and security. My background includes various security principles, with a focus on binary analysis.

I want to explore new research areas in computer systems and programming languages.

SKILLS

Programming: C, C++, Python, Javascript, Golang, Rust

Technologies: Linux, Docker, LLVM

Languages:

Vietnamese	Native	
English	IELTS 7.5	(C1)
Japanese	JLPT N2	(B2)

EDUCATION

2025 - Present	M.S., Computer Science	Northwestern University
2016 - 2020	Bachelor of Engineering, Computer Science	Ho Chi Minh University of Technology

WORK EXPERIENCES

9/2019 – Present	Senior Security Engineer	BShield - Verichains, VNG
• Mobile security for Android and iOS applications • Code obfuscation using LLVM • Code obfuscation for Mach-O binary • Performed audits for Vietnam Citizen Card • Performed audits for blockchain projects • Performed audits for cryptographic libraries BShield / BShield Secure-ID / TSShock		
6/2019 – 8/2019	Malware analyst intern	Viettel Cyber Security
• Malware analysis • Malware removal and prevention		

PROJECTS

Gateway Plugins 2024	BShield API Protection Plugins	
• Implement BShield API Protection plugins for a variety of frameworks, including: NGINX (C++); Apache (C++); Kong (Golang); HAProxy (Lua); APISIX (Lua); Spring/Websphere (Java); Express (Javascript); and Kubernetes Ingress;		
Linux 2024	BShield Box	
• Package BShield solution in a secure Linux machine (physical) • Re-implement as a QEMU-based customized Linux system • Bootstrap customized Linux system with unique init process, and system components • Using: LUKS+TPM2, udev, Docker		
Cryptography 2023	TSShock: Private key recovery for libraries implementing Threshold ECDSA based on GG18	tsshock
• Implemented attacks in multiple libraries implementing GG18 • Perform brute-force search in parallel using CUDA • A basic understanding of multiparty computation (MPC) and Zero-Knowledge Proofs (ZKP)		
NFC security 2023	Vietnam Citizen Card security audits	
• In-depth understanding of the ICAO 9303 protocol and PACE protocol • Simulate the protocol for testing of software and hardware • Create counterfeit cards through simulation • Bypass security of multiple software and hardware		

Obfuscation 2019-2023	Simulating Loader for Mach-O Binary Obfuscation and Hooking • Thorough understanding of the Mach-O binary format, the loader dyld and Objective-C runtime • Understand basic and advanced Mach-O obfuscation techniques • Created a new method for API obfuscation in Mach-O binaries	preprint / code
Obfuscation 2021-2022	Source code obfuscation using LLVM • Use Obfuscator-LLVM • Update obfuscation passes to LLVM 14 and port them to New Pass Manager • Integrated with Golang using gollvm, Web Assembly using emscripten, and Rust	
Memory Forensics 2019-2023	Live memory forensics without memory extraction • Understand Windows memory model • Familiar with memory forensic techniques and malware techniques • Apply memory forensics directly on virtual memory through a kernel driver • Controlling of kernel drivers using IOCTL with a user space process • Implemented and modified Pool Tag Quick Scanning to run in virtual memory	dissertation / paper

AWARDS AND HONORS

I usually participated in Capture the Flag (CTF) competitions during my undergraduate years. Below is the list of competitions I participated in with members of my university club or with the Vietnamese security community.

2021	Google Capture The Flag (Team)	19 th
2020	Google Capture The Flag (Team)	11 th
2020	International Olympiad in Cryptography (Team)	2 nd Prize
2019	ASEAN Student Contest on Information Security (Team)	Finalist
2019	VIETTEL MATES CTF (Team)	2 nd
2018	ASEAN Student Contest on Information Security Qualify Round (Team)	3 rd
2018	VIETTEL MATES CTF (Team)	Finalist
2018	VNPT SecAthon (Team)	3 rd

SERVICES

2022 - Present	Bachelor's thesis co-advisor My former advisor, Dr. Khuong Nguyen-An, invites me to be a co-advisor for his students' bachelor's thesis. • Nguyen Thien Nhan, 2024 - 2025, Emulate Smart Card with Proxmark 3 • Pham Nguyen Nam, 2024 - 2025, Static Binary Repairing with Code Insertion • Do Dinh Phu Quy, 2023 - 2024, Sandboxing Powershell scripts for ransomware detection • Vo Van Tien Dung, 2022 - 2023, Windows Memory Forensics: Detecting hidden injected code in a process
2019 - 2020	Efiens Cyber Security Club Former Leader Efiens is a highly advanced student group in reverse engineering, cryptography, pwning and web exploitation. See our competition achievement. Our members are Flare-on, Olympiad in Cryptography winners, and National cyber security competition winners.

TALKS

Black Hat MEA Saudi 2024	Critical Zero-Knowledge Proof Forgery Attack on zkEVM On behalf of the Verichains Cryptography team: Anh Khoa Nguyen , Thanh Nguyen	conference
VXCON HongKong 2024	Stopping the New Wave of Mobile Banking Malware infecting users in South East Asia On behalf of the BShield team: Anh Khoa Nguyen , Thanh Nguyen	conference / video
Black Hat USA 2023	TSSHOCK: Breaking MPC Wallets and Digital Custodians for \$BILLION\$ Profit Duy Hieu Nguyen, Anh Khoa Nguyen , Huu Giap Nguyen, Thanh Nguyen, Anh Quynh Nguyen	conference / video
Hack In The Box Phuket 2023	TSSHOCK: Breaking MPC Wallets and Digital Custodians for \$BILLION\$ Profit Duy Hieu Nguyen, Anh Khoa Nguyen , Huu Giap Nguyen, Thanh Nguyen, Anh Quynh Nguyen	conference / video

PUBLICATIONS

Preprint	Simulating Loader for Mach-O Binary Obfuscation and Hooking Anh Khoa Nguyen , Thien Nhan Nguyen	pdf
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FDSE 2024

Live Memory Forensics on Virtual Memory
Khoa A. Nguyen, Tien-Dung Vo-Van, Anh-Quynh Nguyen, Thanh Nguyen-Le, Dinh-Thuan Le, and Khuong Nguyen-An

pdf

Whitepaper 2023

New Key Extraction Attacks on Threshold ECDSA Implementations
Duy Hieu Nguyen, **Anh Khoa Nguyen**, Huu Giap Nguyen, Thanh Nguyen, Anh Quynh Nguyen

pdf

REFERENCES

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